
Flashlight Project Proposal

Ahmad Taka

ahmadtak@mit.edu

MIT EECS

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1 Summary

For my final project, I will design and build a compact, thermally-managed boost LED driver for a Cree XHP50.3 6V LED (XHP50D-00-0000-0D0UH240G). The goal is to create a high-powered flashlight that is practical for everyday use, bright enough to light up a workspace or driveway, but efficient and cool enough to hold comfortably. I've always liked the idea of having a personal, custom-built light that's powerful yet portable, and I want to design the power electronics myself rather than rely on off-the-shelf flashlight drivers.

Over the summer, I prototyped the lipo and controller on an STM32 dev board I designed. It can charge a Li-ion cell, supply up to 3 A on a Power Path rail (Built using the BQ2125), and read an onboard accelerometer that I used to turn on an LED with a simple "shake-to-wake" gesture. I also tested several commercial boost drivers, including the TPS61169 and TPS61500, but they failed to deliver enough current to reach full LED brightness. That experience pushed me toward developing my own higher-power, constant-current driver architecture. The new design uses a boost converter that measures LED current through a shunt resistor and regulates it directly for consistent brightness and thermal performance. Initially for this project, it will run from 3 V to 4.3 V rail from a bench top power supply. The battery and digital control integration planned as a later extension once the converter stage is proven stable and thermally efficient.

For the graduate extension of the project, I plan to use Tellegen's theorem to track how energy moves through the converter and confirm that the measured power losses match the theory. This approach will help ensure that every part of the circuit, from the switches to the inductor contributes predictably to the total energy flow, making it easier to identify where thermal improvements can be made. I also want to build a small switched capacitor voltage doubler to create short, high-voltage flashes on the LED and study how these quick bursts affect brightness and heat (Building on the work in Lab 3).

2 Hardware and System Diagram

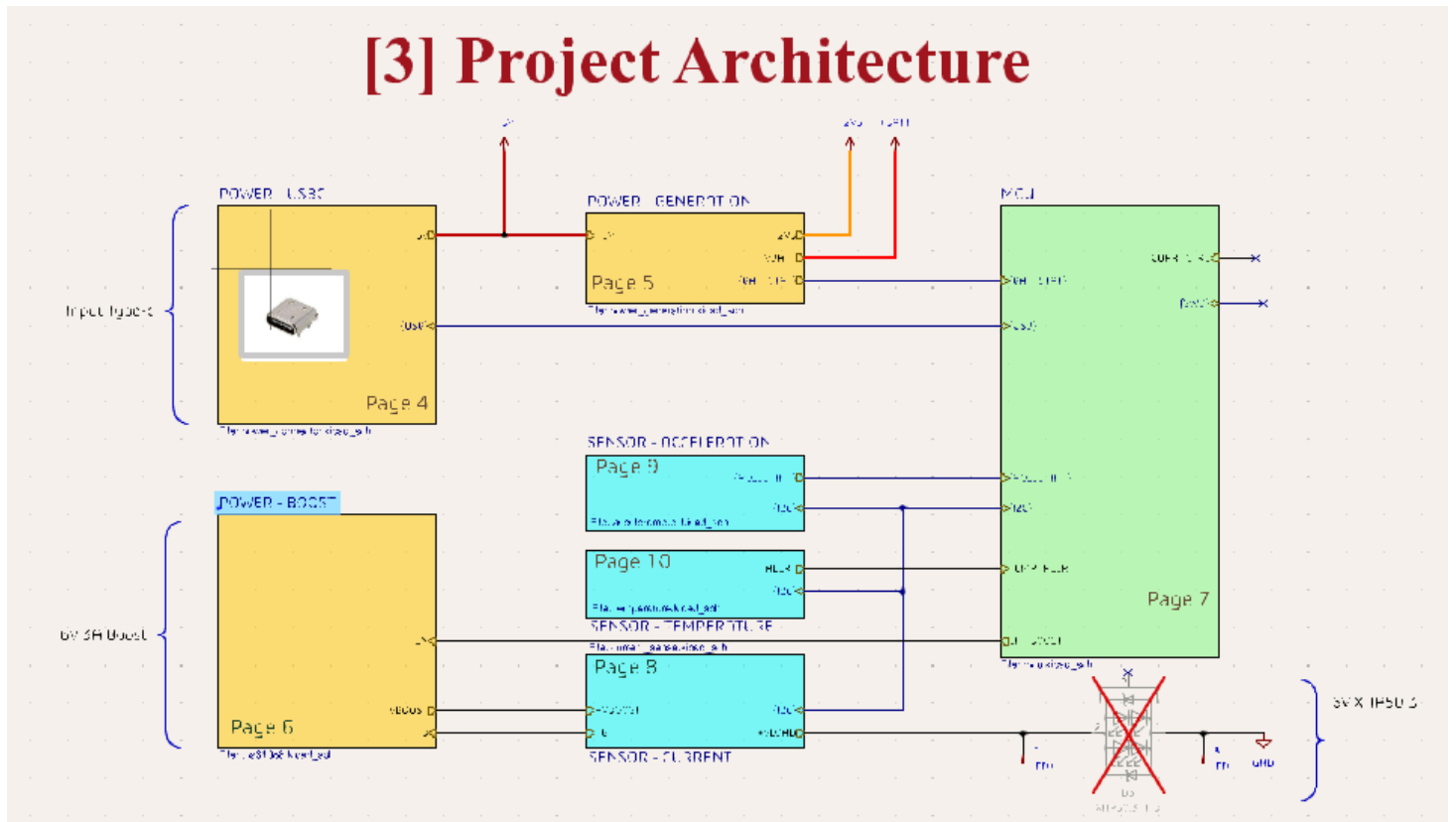


Figure 1: Block diagram of the flashlight system, showing power and control subsystems.

This block diagram outlines the overall flashlight system, including power input, sensing, and control subsystems. For the final project, the primary focus is on the Boost Converter block, shown as the section responsible for stepping up the voltage and providing constant-current regulation to drive the Cree XHP50.3 LED. The converter senses output current through a shunt resistor and feeds this information back to maintain stable LED brightness across varying input voltages.

This design is an initial draft architecture, where only the boost stage will be developed, tested, and characterized within the current project scope. Future integration with the STM32-based logic, sensors, and battery management circuitry will come later. A key part of this phase is evaluating whether GaN FETs or traditional silicon MOSFETs offer better performance in terms of thermal handling, switching frequency, and overall circuit footprint for the synchronous boost topology.

2.1 Initial Boost Design with Feedback

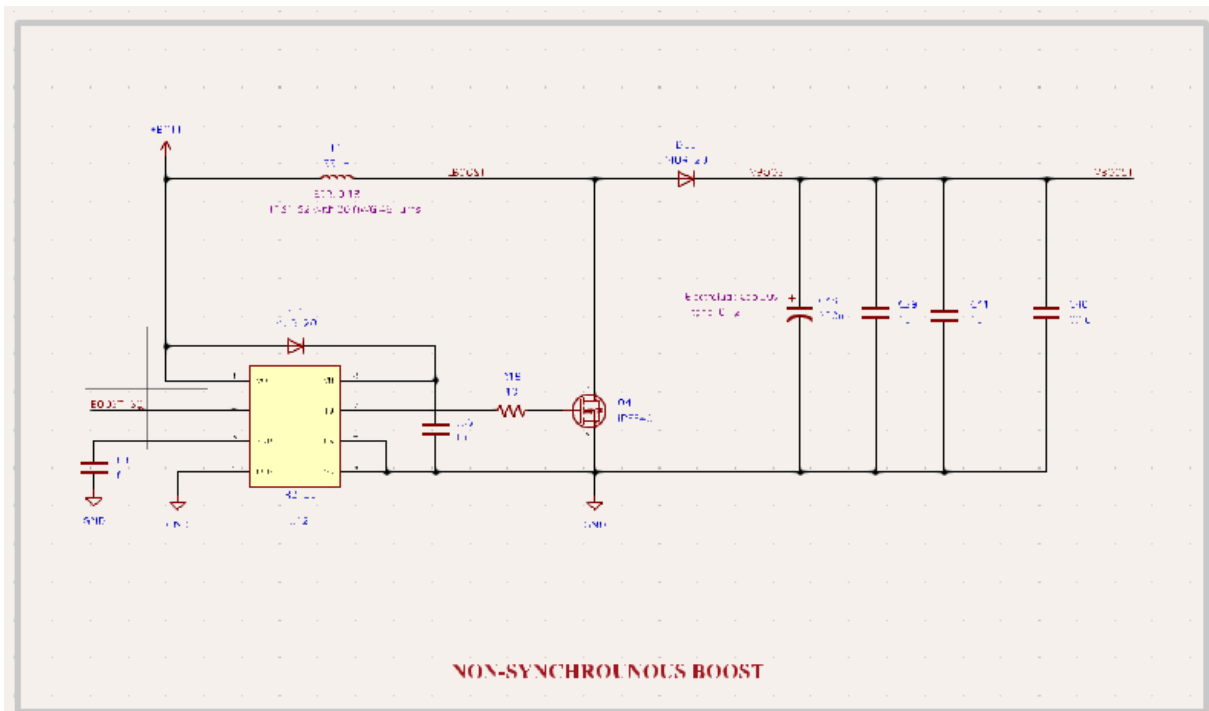


Figure 2: Schematic of the synchronous boost converter.

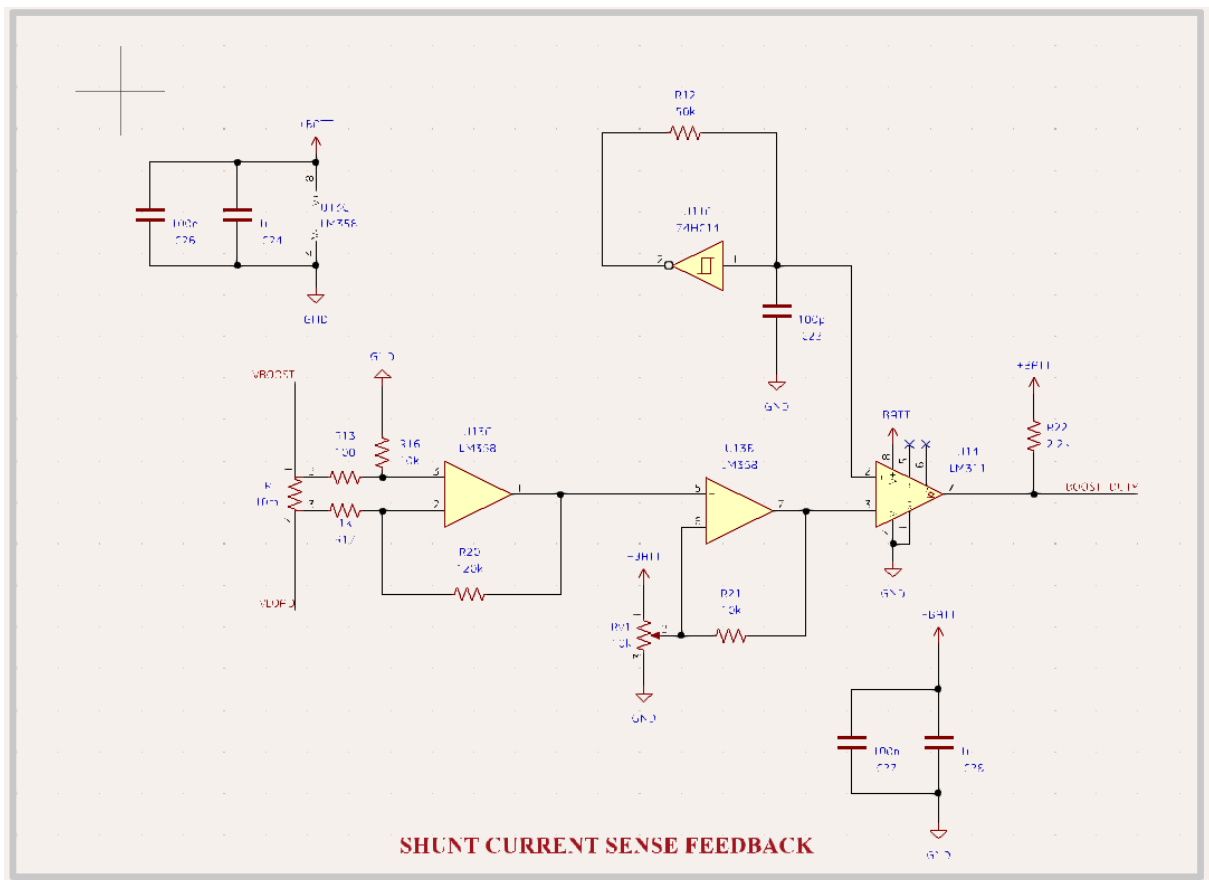


Figure 3: Schematic of the current sensing and feedback network.

This initial design uses a version of the boost converter in Lab 2, a high-current, non-synchronous boost converter, as the main power stage that steps the input voltage (3–4.3 V) up above the 5.6 V (forward-voltage) required by the Cree XHP50.3 LED. However unlike the lab, the converter operates in current-mode control. The LED current is sensed through a high-side shunt resistor, whose voltage drop represents the real-time LED current. The high-side placement of the shunt allows for accurate current measurement without interfering with the LED ground reference.

That sense voltage feeds into a dual op-amp circuit configured as a differential amplifier and comparator. The amplifier conditions the shunt voltage and compares it against a reference voltage generated by a potentiometer. By adjusting the potentiometer, the reference voltage changes, effectively controlling the output current setpoint. The op-amp's output is routed through a variable duty cycle control circuitry where the duty cycle can range from 0.5 to 0.85 to maintain the desired LED current.

In summary, the system forms a closed-loop constant-current boost converter, where the boost stage supplies the main power conversion, and the op-amp plus digital potentiometer network provides adjustable current regulation.

2.2 GaN Vs Traditional FETs

GaN lets you push $3\times$ higher switching frequency, which cuts L or output cap size by $3\times$ and lowers total FET heat at this power. Silicon at lower f_{SW} pays more in switching and gate-drive loss, so even with similar RDS(on), total FET loss is higher. Although not a fair comparison the table below shows how a QFN GaN Chip (EPC2302) stacks up against a larger, older Si FET (IRF540N) in this application. Estimates below isolate the power switches only and ignores how the gate is driven.

Parameter	Si FET (IRF540N)	GaN FET (EPC2302)
f _{SW} (MHz)	0.4	1.2
L (μH), V _{in} =3.3V	2.27	0.76
RDS(on) (mΩ)	44	3.8
Q _g · V _g (nC · V)	630	40
Switching (W)	0.79	0.12
Gate-drive loss (W)	0.50	0.10
Total loss (W)	1.29	0.22
Size note	Baseline	$3\times$ smaller L/C

Table 1: Table comparing estimated losses and component sizes between Silicon and GaN FETs in the boost converter application.

3 Special Components and BOM

TC4426 High-speed Low side driver for the GaN FET.

Current-sense shunt $R1 = 10\text{ m}\Omega$ Sets LED current for the op-amp loop. Requires Kelvin sense.

Part	Quantity	Price \$
TC4426 (Sync Boost IC, U6)	1	2.20
Shunt Resistor $10\text{ m}\Omega$ (R1)	1	0.54
EPC2302 eGaN FET	10	34.58
INA180A1 (Current-Shunt Amp, U8)	1	-
Cree XHP50.3 HD 6 V LED (D3)	5	37.95

Table 2: Bill of materials excerpt highlighting the essential boost IC, sense shunt, GaN option (EPC2302), and notable system parts used in the flashlight driver

4 Project Scope and Management

For this project, I will focus on designing, building, and testing the boost converter stage with constant-current feedback to drive the Cree XHP50.3 LED. I will also design two other supplies to explore the use of alternate power converters. The first is the same boost topology however with GaN FETs to compare performance and thermal characteristics against the traditional silicon FET design. The second is a small switched-capacitor voltage doubler to create high-voltage pulses for the LED, building on the switched-capacitor concepts from Lab 3.

In the first phase, I'll build a basic prototype using the Totem Board from Lab 4 and a 250 kHz switching circuit with current feedback, implemented entirely in analog. This setup will build directly on the work from Lab 3, where I developed and tested the fundamental switching and sensing stages. The goal is to verify that the constant-current feedback loop operates as expected before integrating the full boost converter. Once the analog prototype is stable, I'll measure temperature rise and efficiency across different loads and input voltages to guide the final design.

In the second phase, I'll build a GaN-based version of the boost converter to compare performance. Using GaN FETs should allow higher switching frequencies, reducing inductor/capacitor size and improving thermal performance. I'll follow a similar testing procedure to evaluate efficiency and heat dissipation.

Finally, I'll design and test a switched-capacitor voltage doubler to create short, high-voltage pulses for the LED. This stage will explore how pulsed operation affects brightness and thermal characteristics, building on the switched-capacitor concepts from Lab 3.

5 Timetable

- **Week starting Nov 10:** Lock in Schematic and design choices. Finalize BOM and order parts (DONE).
- **Week starting Nov 17:** Began modeling and circuit construction (IN PROGRESS).
- **Week starting Nov 24:** Design mechanical enclosure. Make GaN version and switch-cap version.
- **Week starting Dec 1:** Test Boards and integrate into system.

- **Week starting Dec 8:** Show off flashlight.